



AttentiveMOS: A Lightweight Attention-Only Model for Speech Quality Prediction

*Authors: Imran E Kibria & Donald S. Williamson
Dept. of Computer Science & Engineering, The Ohio State University.*



THE OHIO STATE UNIVERSITY

Modeling Speech Quality - Key Challenges



Mean Opinion Score (MOS)

MOS is a 5-point scale (1:worst, 5:best) for assessing perceptual speech quality.



Oversized Networks

SOTA models are too large for real-world applications & for aiding downstream tasks.



Subjective Variance

Training models is challenging as listener ratings can vary for the same speech signal.



Key Insights

1

Attention-based

Architecture design inspired by the success of attention in language¹, vision², & speech³

2

Compact

Small token size leads to compact models. Deeper, not wider models, are preferable.

3

Variance Aware

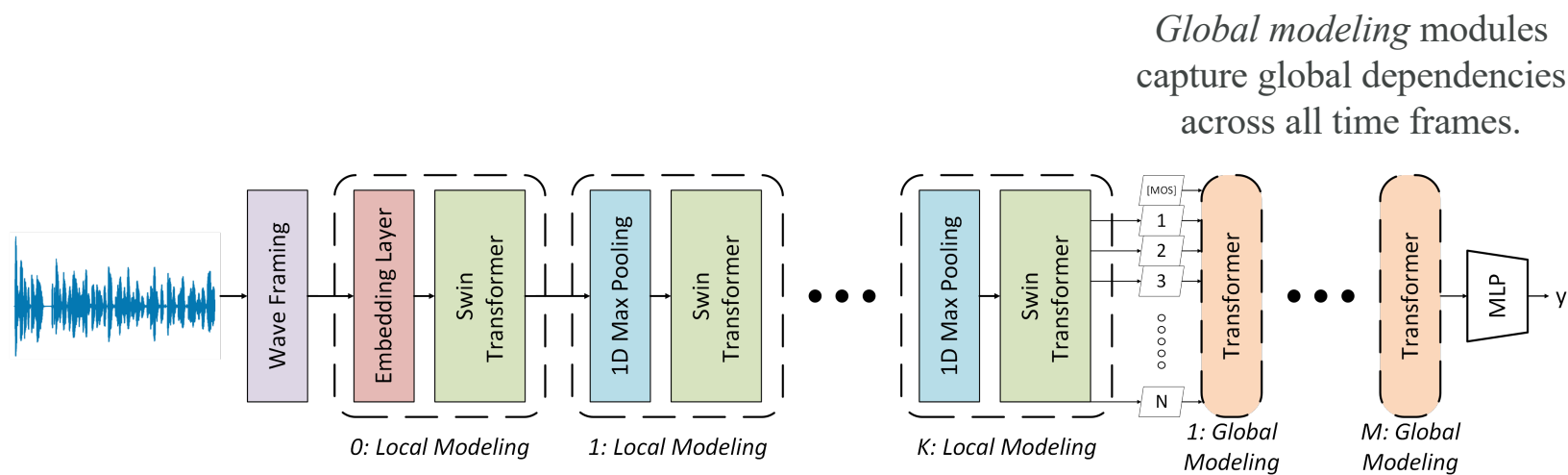
Utterance labels with lower variance in listener ratings are more reliable.



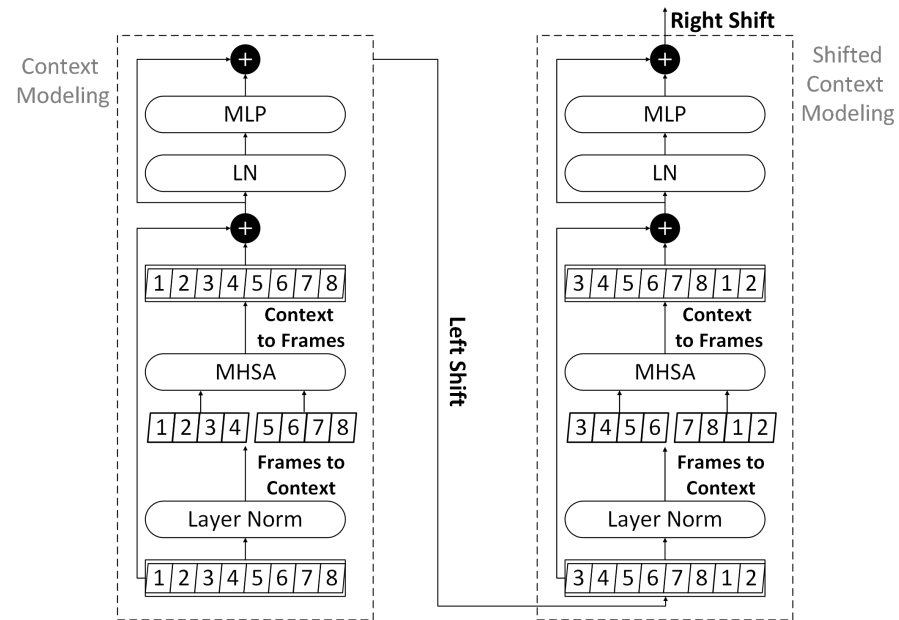
Small Attention-driven Model Architecture



Attentive-MOS



Swin Transformer ⁴



Training Strategy: Tackling Label Variance



Proposed Objective

Assign higher importance to samples with smaller variance in ratings.

$$l = \ln \left(1 + \frac{|y - \mu|}{\sigma + \varepsilon} \right)$$

y MOS prediction

μ MOS label

σ Standard deviation in utterance ratings

ε A small positive constant

Sequential Self-Teaching

- A series of models are trained sequentially.
- Teacher & student models share the same architecture.

$$\mu_m = (\alpha_o \cdot \mu) + \sum_{i=1}^m (\alpha_i \cdot y_{i-1})$$

- μ_m Student label in stage m
- μ MOS label
- y_i Model prediction from stage i
- α_i label weight.
Sum of all weights must sum to 1

Ablation Studies & Performance Outcomes



Datasets

	Dataset	Number of Samples			Min. listener ratings per sample	Standard deviation in ratings
		Training	Validation	Testing		
<i>synthetic</i>	Samsung Open Mean Opinion Score (SOMOS) ⁵	14,000	3,000	3,000	17	1.124
	2022 Blizzard Voice Conversion Challenge (BVCC '22) ⁶	4,974	1,066	1,066	8	--
<i>non-synthetic</i>	Non-Intrusive Speech Quality Assessment (NISQA) ⁷	11,020	2,700	952	5	0.666

Component Analysis

- Acoustic content of speech is independent of sequential dependencies.
- Our proposed objective requires further study.

		Clean Set		Full Set	
		MSE ↓	PCC ↑	MSE ↓	PCC ↑
Positional Encoding	Included	0.292	0.344	0.140	0.318
	Excluded	0.265	0.422	0.132	0.394
Frame Compression	Linear	0.304	0.289	0.148	0.256
	Avg Pooling	0.292	0.347	0.138	0.336
	Max Pooling	0.265	0.422	0.132	0.394
Loss Function	MAE	0.267	0.408	0.128	0.397
	MSE	0.257	0.449	0.127	0.412
	Proposed	0.265	0.422	0.132	0.394



Accuracy & Generalization

Model	Parameters	Training Set	MSE ↓	PCC ↑	SRCC ↑
MOSNet ⁸	~500K	VCC2018	0.843	-0.075	-0.084
		SOMOS-full	0.598	0.218	0.238
		SOMOS-clean	0.729	0.352	0.347
LDNet ⁹	~0.96M	BVCC	1.011	0.040	0.032
		SOMOS-full	0.581	0.262	0.275
		SOMOS-clean	0.642	0.397	0.386
SSL-MOS ¹⁰	~95M	BVCC	2.217	0.229	0.230
		SOMOS-full	0.564	0.296	0.313
		SOMOS-clean	0.625	0.453	0.444
AttentiveMOS	~86K	BVCC	0.390	0.113	0.109
		SOMOS-full	0.312	0.445	0.433
		SOMOS-clean	0.257	0.449	0.442



Sequential Self-Teaching Analysis

- In each successive stage, student model achieves greater performance.¹¹

Stage	Validation Set		Test Set	
	MSE ↓	PCC ↑	MSE ↓	PCC ↑
m = 0	0.127	0.412	0.126	0.403
m = 1	0.126	0.409	0.123	0.414
m = 2	0.123	0.431	0.121	0.432
m = 3	0.124	0.428	0.121	0.429



Key Takeaways

1

Speech Acoustics

Without positional information or max-pooling, higher accuracy is achieved.

2

Generalizability

SSL-MOS & AttentiveMOS are similar on in-domain sets; SSL-MOS can outperform on out-of-domain datasets.

3

SUSTAIN

Sequential self-teaching strategy is proven to improve model performance on noisy label datasets.



Conclusion & Future Work



Final Remarks

- ✓ AttentiveMOS: a compact, attention-only model for subjective speech quality assessment.
- ✓ Outperforms small models and matches large models on in-domain data.
- ✓ Sequential self-teaching boosts performance under noisy labels.
- ✓ Future work: enhance generalization across out-of-domain datasets.

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Thank you!

Contact:
kibria.5@osu.edu,
williamson.413@osu.edu

